Player 2 - Field Manual

> **Bomb Diffusal Activity.** Player 2 operates the two RGB LEDs, the four coloured buttons (Red, Yellow, Green, Blue) and the rotary encoder. Player 1 reads the screen to you; you look things up here and call out what to do. **Do not let Player 1 see this manual.**

You will also need the physical tools tucked into this manual:

- a printed **ruler** marked in the map's scale, and
- a printed **protractor** (0-360 degrees).

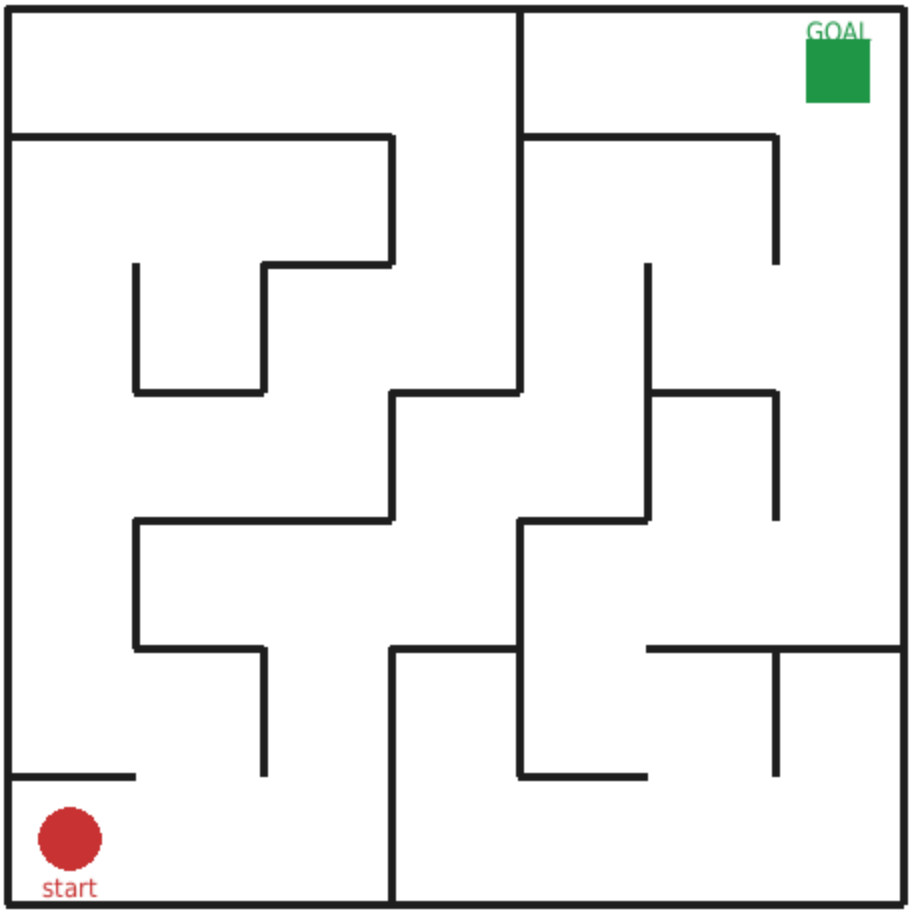
> *[PLACEHOLDER: print the ruler and protractor on transparent film and attach them to the inside cover. The ruler's scale must match the map's 1-mile bar.]*

Section 1 - Mazes (Puzzle 4: Invisible Maze)

Player 1's screen shows a dot and a goal but **no walls**, and a label **“Maze 1/2/3”**. Find the matching maze below. The red dot is the start; the green square is the goal. Call out each move (up/down/left/right). Player 1 cannot move through a wall.

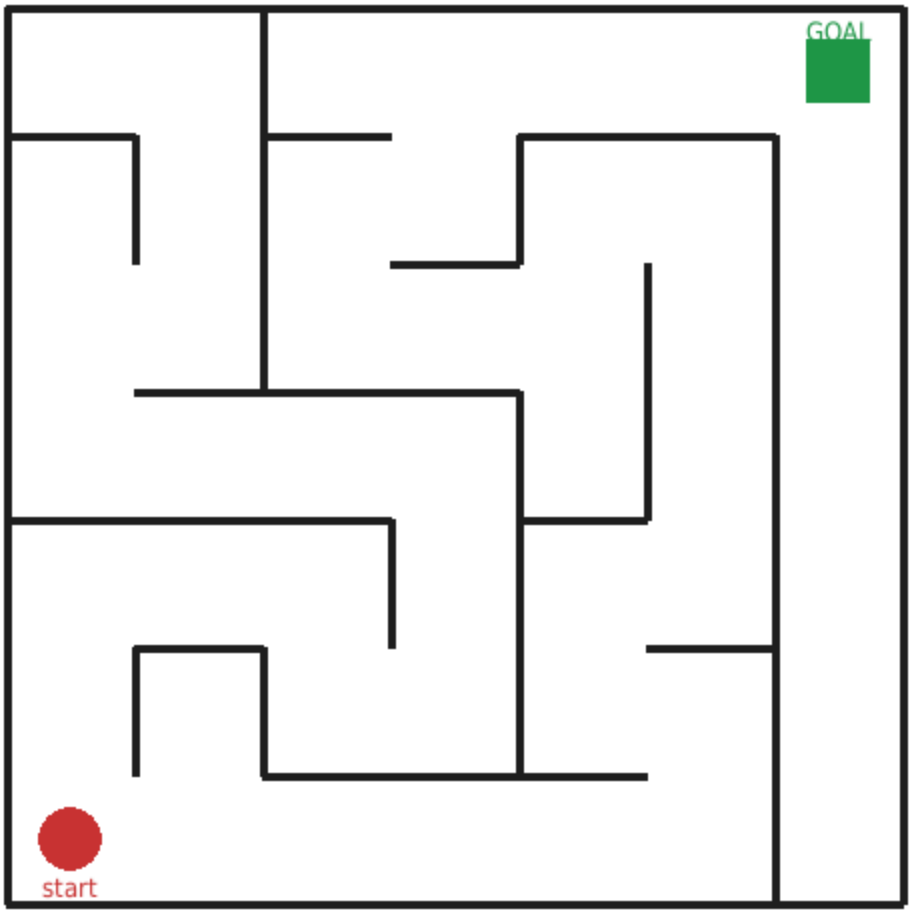
Maze 1

Maze 1



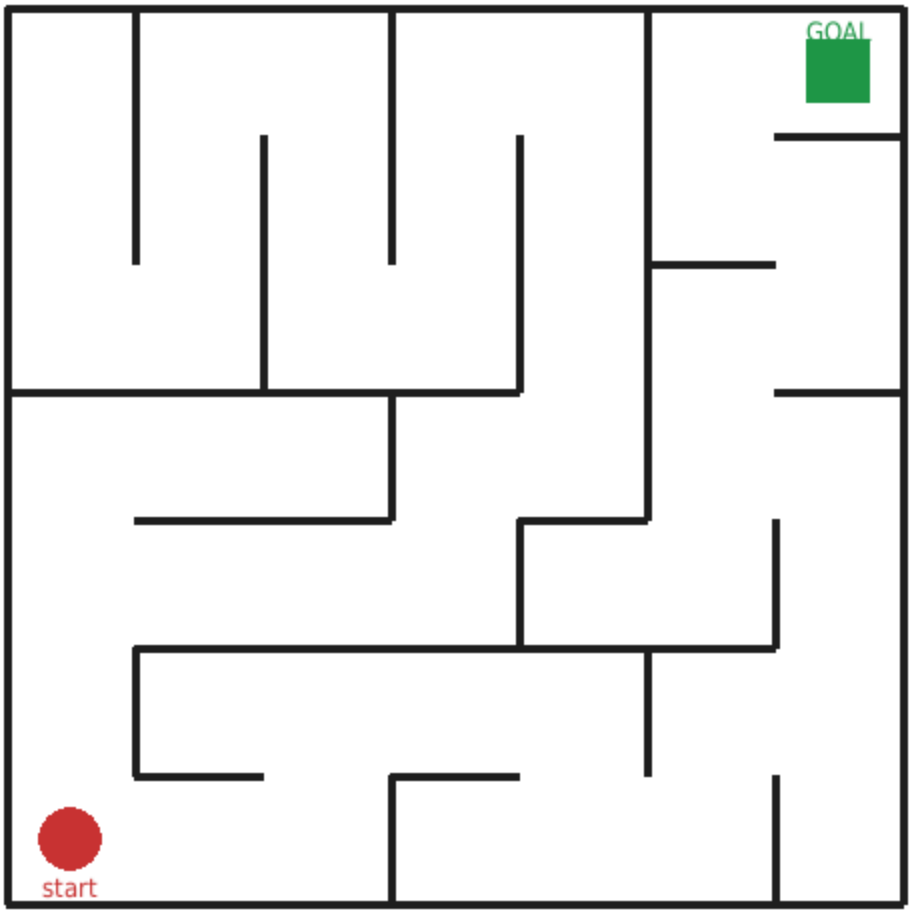
Maze 2

Maze 2



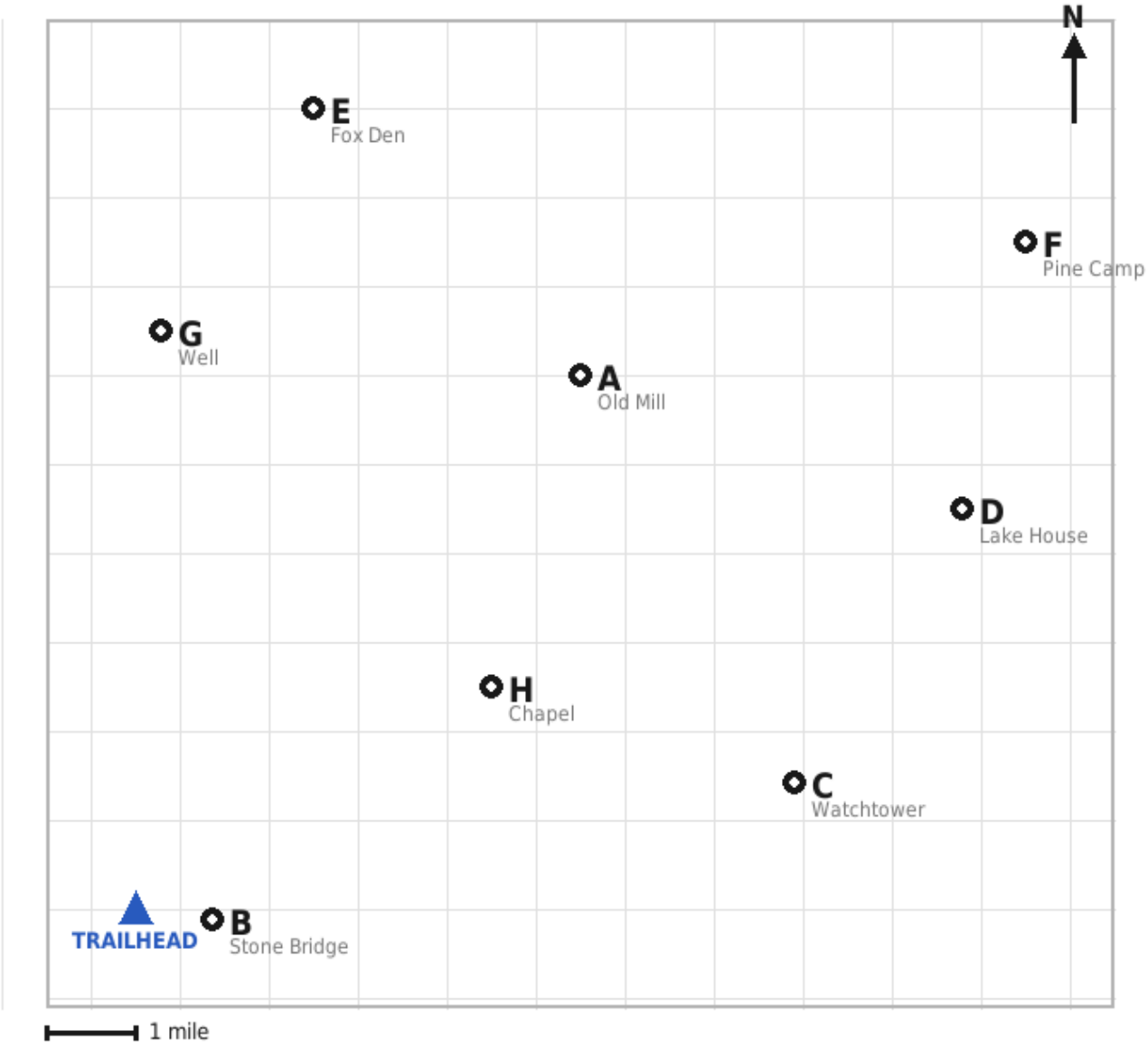
Maze 3

Maze 3



Section 2 - Trail Map (Puzzle 5: Navigation)

Player 1 reads you a list of directions from the **trailhead** (blue triangle). Use the ruler and protractor to trace the route on this map, then tell Player 1 the **letter** of the landmark where the trail ends.



How to measure: bearings are compass degrees (0 deg = North, measured clockwise). Place the protractor's centre on your current point, set 0 deg to North, and turn by the stated angle. Measure distance with the ruler against the 1-mile scale bar.

Landmarks:

Letter	Landmark
A	Old Mill
B	Stone Bridge
C	Watchtower
D	Lake House
E	Fox Den

Letter	Landmark
F	Pine Camp
G	Well
H	Chapel

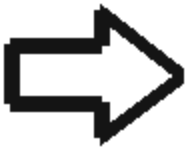
Section 3 - Symbol Traits (Puzzle 8: Symbols Lock)

The screen shows five symbols; **four share one hidden trait** and one does not. Player 1 names the symbols (see their Symbol Names card); look each one up below and find the trait four of them share. The **odd one out** is the answer Player 1 clicks.

The hidden trait is always one of these:

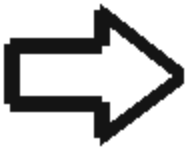
- Same number of strokes
- All curved (no straight edges)
- All symmetrical (have a mirror line)
- All closed shapes (fully enclose an area)

Symbol	Name	Strokes	Curved	Symmetrical	Closed
	Spiral	1	yes	no	no
	Circle	1	yes	yes	yes
	Crescent	2	yes	no	yes

Symbol	Name	Strokes	Curved	Symmetrical	Closed
	Square	4	no	yes	yes
	Triangle	3	no	yes	yes
	Pentagon	5	no	yes	yes
	Star	5	no	yes	yes
	Heart	3	yes	yes	yes
	Arrow	3	no	yes	no
	Plus	3	no	yes	no

Section 4 - Symbol Values (Puzzle 9: Glyph Addition)

The screen shows an equation made of symbols. Player 1 reads it out; look up each symbol's number here, do the arithmetic, and give Player 1 the total to type.

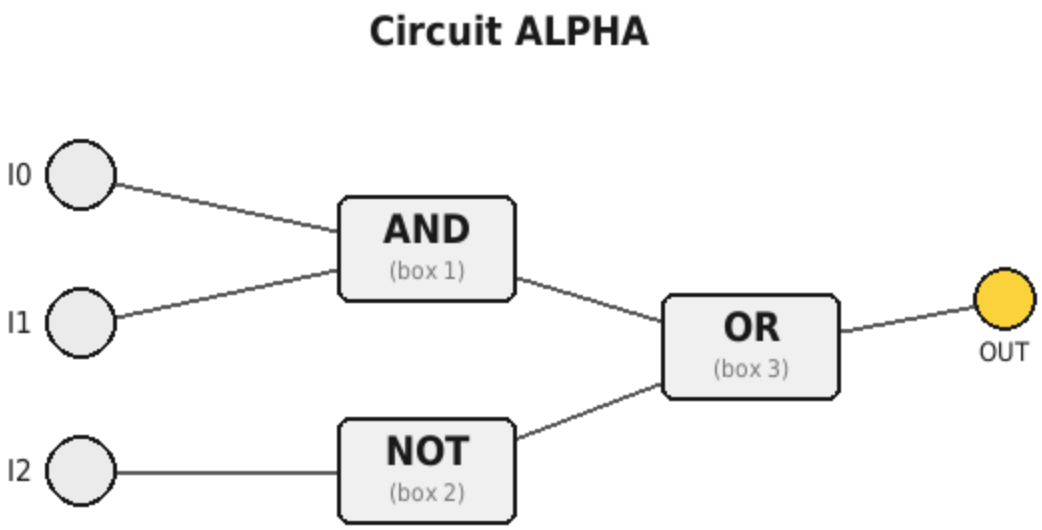
Symbol	Name	Value		Symbol	Name	Value
	Spiral	1			Circle	2
	Crescent	3			Square	4
	Triangle	5			Pentagon	6
	Star	7			Heart	8
	Arrow	9			Plus	10

Section 5 - Circuit Diagrams (Puzzle 6: Turn on the LED)

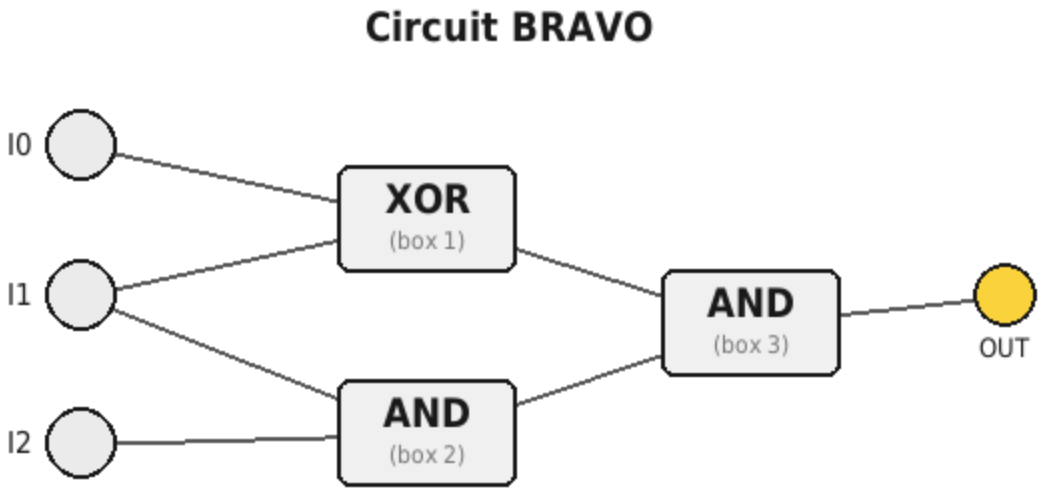
Player 1's screen shows the same circuit but with the gate boxes **numbered and blank**, plus a label **“Circuit ALPHA/BRAVO/CICERO”**. Find the matching diagram below - it tells you which gate each numbered box is. Work out which input switches (0/1) make the output lamp light, and tell Player 1 which switches to toggle.

Gate reminders: **AND** = on only if both inputs on. **OR** = on if either input on. **XOR** = on if the inputs differ. **NOT** = flips its one input.

Circuit ALPHA



Circuit BRAVO



Circuit CICERO

Circuit CICERO

